

YU FENG

Ph.D. Candidate | Imaging Systems

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PROFESSIONAL SUMMARY

Trilingual researcher (English, Japanese, Chinese) with 5+ years of experience in LSI design, hardware/software implementation, and performance characterization. Specializes in multi-tap CMOS Image Sensor (CIS) systems development, achieving 56-126 dB single-exposure HDR with motion/flicker mitigation. Additionally developed robust biomedical imaging systems resilient to ambient light and motion artifacts. Proven track record of full-cycle tape-out process (180 nm), FPGA prototyping, and first-author peer-reviewed publications.

TECHNICAL SKILLS

IC Design & Verification: Cadence Virtuoso (Schematic/Layout), Spectre (Simulation), Calibre (DRC/LVS), PCB design, full-cycle tape-out (180 nm CIS process).

Characterization: Dynamic Range (DR)/SNR analysis, Linearity tuning, Full Well Capacity (FWC) optimization, Crosstalk suppression.

Hardware & Architecture: Multi-tap CIS architecture, FPGA (Verilog, Quartus, ModelSim).

Software: MATLAB (Image processing, simulation), Python, C/C++.

Languages: English (TOEIC 990/990), Japanese (JLPT N1), Chinese (Native).

EDUCATION

Ph.D. Shizuoka University Hamamatsu, Japan Nanovision Technology Advisor: Prof. Keiichiro Kagawa	2023.10 – 2026.09 (Expected)
M.Eng. Shizuoka University Hamamatsu, Japan Electronics Engineering	2021.10 – 2023.09
B.Eng. Shizuoka University Hamamatsu, Japan Electrical and Electronic Engineering	2016.04 – 2020.03

CORE RESEARCH EXPERIENCE

High Dynamic Range (HDR) Imaging System Development	2023 – Present
- Systematically optimized reset and transfer gate voltages to suppress residual charge crosstalk, achieving a programmable 56–126 dB Single-Exposure HDR. - Developed a programmable charge-splitting method that mitigates LED flicker and motion artifacts by implementing a repeated or a random shutter sequence to suppress motion artifacts and decouple exposure from the pulsed light source. - Utilized a 4-tap CIS to extend dynamic range in three discrete steps, maintaining a transition SNR >30 dB across the full range for signal fidelity in automotive applications.	
Ambient-Light-Robust Motion-Resistant Biomedical Imaging System Development	2021 – 2023
- Improved ambient light tolerance by >10x in clinical environments by engineering an 8-tap CIS system that performs quantitative tissue imaging synchronized with pulsed illumination. - Led the LSI design and full tape-out process of a multi-aperture 4-tap CIS, resulting in a successfully fabricated prototype (2024) for non-invasive multi-spectral biomedical imaging.	

PUBLICATIONS

1. Performance Characterization and Tuning of a Charge-Splitting High Dynamic Range 4-Tap CMOS Image Sensor Yu Feng, et al., <i>Sensors</i> , DOI: 10.3390/s25226953	2025.11
2. Programmable Dynamic Range HDR Imaging with LED-Flicker and Motion Artifact Mitigation Using a Four-Tap CMOS Image Sensor Yu Feng, et al., <i>IEEE Sensors Journal</i> , DOI: 10.1109/JSEN.2025.3557801	2025.03
3. Motion-Resistant Three-Wavelength Spatial Frequency Domain Imaging System with Ambient Light Suppression Using an 8-Tap CMOS Image Sensor Yu Feng, et al., <i>Journal of Biomedical Optics</i> , DOI: 10.1117/1.JBO.29.1.016006	2024.01
4. Spatial Frequency Domain Imaging System Using a Scanning Micro-Mirror Kenta Nakazawa, Yu Feng, et al., <i>Sensors and Actuators A: Physical</i> , DOI: 10.1016/j.sna.2025.116421	2025.03
5. Resolving Multi-Path Interference in Compressive Time-of-Flight Depth Imaging with a Multi-Tap Macro-Pixel Computational CMOS Image Sensor Horio Masaya, Yu Feng, et al., <i>Sensors</i> , DOI: 10.3390/s22072442	2023.01

RECENT INTERNATIONAL CONFERENCE PRESENTATIONS

1. Room-Light Operation of a Three-Wavelength Spatial Frequency Domain Imaging System Using Pulsed Illumination and an 8-Tap CMOS Image Sensor Yu Feng, et al., <i>European Conference on Biomedical Optics 2025</i> , Munich, Germany.	2025.06
2. Programmable Dynamic Range Extension up to 110 dB Based on Charge-Splitting Method with 4-Tap CMOS Image Sensor Yu Feng, et al., <i>International Image Sensor Workshop 2025</i> , Hyogo, Japan.	2025.06
3. Multi-Tap CMOS Image Sensor with Programmable Functional Exposure: Application to Structured Light Based Quantitative Tissue Imaging Yu Feng, et al., <i>Optica Imaging Congress 2024</i> , Toulouse, France.	2024.07

INDUSTRY EXPERIENCE

QA Engineer , Meidensha Nagoya, Japan	2020.04 – 2021.09
Ensured reliability of EV motor production lines by implementing rigorous quality assurance protocols and defect analysis, reducing line stoppages.	

AWARDS & HONORS

Graduate School Scholarship , Amano Foundation	2023 – 2026
Outstanding Academic Record , Shizuoka University	2024 – 2026